

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 24

20-08-2026

24. WRITE A PROLOG PROGRAM TO SUGGEST DIETING SYSTEM BASED ON DISEASE.

AIM

To write and execute a Prolog-based expert system that suggests a suitable diet based on a given disease. The program demonstrates the use of knowledge representation, facts, rules, and logical queries for developing a simple expert system.

ALGORITHM

1. Start the program.
2. Define different diseases as facts.
3. Define suitable diet recommendations for each disease.
4. Create a predicate to associate a disease with its recommended diet.
5. Accept a disease as a query.
6. Search the knowledge base for the corresponding recommendation.
7. Display the suggested diet.
8. Stop the program.

PROLOG PROGRAM

% Write a Prolog program to suggest dieting system based on disease.

diet(diabetes, 'Low sugar and high fiber diet').

diet(hypertension, 'Low salt and low fat diet').

diet(anemia, 'Iron rich foods such as spinach and beans').

diet(obesity, 'Low calorie diet with fruits and vegetables').

diet(ulcer, 'Light food and avoidance of spicy foods').

suggest_diet(Disease, Diet) :-

diet(Disease, Diet).

SAMPLE INPUT

?- suggest_diet(diabetes, Diet).

SAMPLE OUTPUT

Diet = 'Low sugar and high fiber diet'.

RESULT

Thus, the Prolog-based dieting expert system was successfully implemented. The program successfully provided the predefined dietary recommendation corresponding to the given disease.